

From Prime Hecke Non-Concentration to Angular Kummer Collision Decay

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Abstract

Paper 10 separates fixed- ℓ Kummer row-balance from the high- ℓ analytic seam. This note sharpens that bridge. A constant angular non-concentration hypothesis is sufficient but stronger than necessary. Under a density-floor assumption for the local obstruction weights, ordinary prime Hecke non-concentration already implies the angular-energy lower bound needed by Paper 9, after paying the sharp universal cosine loss $1 - \cos(2\pi/\ell) \geq 8/\ell^2$. Consequently PHNC with exponent C gives normalized Kummer collision decay of size $\exp(-c_\alpha \ell^{-C-2} |\mathcal{Q}|)$. In the range $\ell \leq (\log Q)^B$, this exponent still diverges provided C is fixed. The result is conditional on PHNC and does not prove PHNC, Kummer-BV, KRWS, or Erdős–Graham #203.

1 Setup

Let $\mathcal{Q}$ be a finite prime set, let $x_q \in \mathbb{F}_\ell^r$ be Kummer log rows, and let

$$p_q = \frac{|\Gamma_q|}{q-1}.$$

For $A \in \mathbb{F}_\ell^r \setminus \{0\}$, Paper 9 defines angular energy

$$\mathcal{E}(A) = \sum_{q \in \mathcal{Q}} 2p_q(1-p_q) \left(1 - \cos \frac{2\pi(A \cdot x_q)}{\ell} \right),$$

and

$$\mathcal{E}_* = \min_{A \neq 0} \mathcal{E}(A).$$

Paper 9 proves

$$\frac{\mathcal{C}_\ell(N)}{\varphi(P)^2} \leq (1 - \ell^{-r}) e^{-\mathcal{E}_*}. \tag{1}$$

2 The cosine loss

Lemma 1 (Universal cosine gap). *For every prime $\ell \geq 2$ and every nonzero $t \in \mathbb{F}_\ell$,*

$$1 - \cos \frac{2\pi t}{\ell} \geq \frac{8}{\ell^2}.$$

Proof. For $1 \leq t \leq \ell - 1$, the minimum of $1 - \cos(2\pi t/\ell)$ occurs at $t = 1$ or $t = \ell - 1$. Thus it suffices to show

$$1 - \cos \frac{2\pi}{\ell} \geq \frac{8}{\ell^2}.$$

For $\ell = 2$, both sides equal 2. For $\ell \geq 3$, use

$$1 - \cos u = 2 \sin^2(u/2)$$

with $u = 2\pi/\ell$. Since $0 \leq \pi/\ell \leq \pi/3$ and $\sin x \geq 2x/\pi$ for $0 \leq x \leq \pi/2$,

$$1 - \cos \frac{2\pi}{\ell} = 2 \sin^2 \frac{\pi}{\ell} \geq 2 \left(\frac{2}{\ell} \right)^2 = \frac{8}{\ell^2}.$$

□

3 PHNC to angular energy

Hypothesis 1 (Hyperplane PHNC on $\mathcal{Q}$). *There is a constant $C > 0$ such that, for every nonzero $A \in \mathbb{F}_\ell^r$,*

$$\#\{q \in \mathcal{Q} : A \cdot x_q \neq 0\} \geq \ell^{-C} |\mathcal{Q}|.$$

Theorem 1 (PHNC-to-angular bridge). *Assume [Hypothesis 1](#). If there is $0 < \alpha \leq 1/2$ such that*

$$\alpha \leq p_q \leq 1 - \alpha \quad (q \in \mathcal{Q}),$$

then

$$\mathcal{E}_* \geq 16\alpha(1 - \alpha)\ell^{-C-2} |\mathcal{Q}|.$$

Consequently,

$$\frac{\mathcal{C}_\ell(N)}{\varphi(P)^2} \leq (1 - \ell^{-r}) \exp(-16\alpha(1 - \alpha)\ell^{-C-2} |\mathcal{Q}|).$$

Proof. Fix $A \neq 0$. For each q with $A \cdot x_q \neq 0$, [Lemma 1](#) gives

$$1 - \cos \frac{2\pi(A \cdot x_q)}{\ell} \geq \frac{8}{\ell^2}.$$

The density floor gives

$$2p_q(1 - p_q) \geq 2\alpha(1 - \alpha).$$

Therefore

$$\mathcal{E}(A) \geq \sum_{q: A \cdot x_q \neq 0} 2\alpha(1 - \alpha) \frac{8}{\ell^2}.$$

By [Hypothesis 1](#), the summation set has size at least $\ell^{-C} |\mathcal{Q}|$. Hence

$$\mathcal{E}(A) \geq 16\alpha(1 - \alpha)\ell^{-C-2} |\mathcal{Q}|.$$

This is uniform in $A \neq 0$, so it holds for $\mathcal{E}_*$. Substitution into [\(1\)](#) proves the collision bound. □

4 Weighted form

The density-floor assumption is convenient but not always natural. The exact finite statement is weighted.

Hypothesis 2 (Weighted PHNC). *There is a quantity $W_\ell(\mathcal{Q}) > 0$ such that, for every nonzero $A \in \mathbb{F}_\ell^r$,*

$$\sum_{q: A \cdot x_q \neq 0} p_q(1 - p_q) \geq W_\ell(\mathcal{Q}).$$

Theorem 2 (Weighted PHNC-to-angular bridge). *Assume [Hypothesis 2](#). Then*

$$\mathcal{E}_* \geq \frac{16}{\ell^2} W_\ell(\mathcal{Q}),$$

and therefore

$$\frac{\mathcal{C}_\ell(N)}{\varphi(P)^2} \leq (1 - \ell^{-r}) \exp\left(-\frac{16}{\ell^2} W_\ell(\mathcal{Q})\right).$$

Proof. Repeat the proof of [Theorem 1](#), but do not replace $p_q(1 - p_q)$ by a uniform floor. For all nonzero A ,

$$\mathcal{E}(A) \geq \frac{16}{\ell^2} \sum_{q: A \cdot x_q \neq 0} p_q(1 - p_q) \geq \frac{16}{\ell^2} W_\ell(\mathcal{Q}).$$

Then use [\(1\)](#). □

5 High- ℓ consequence

Specialize to primes $q \leq Q$, $q \equiv 1 \pmod{\ell}$, and write

$$|\mathcal{Q}| \asymp \frac{Q}{\ell \log Q}$$

in the standard prime-number-theorem scale. If

$$\ell \leq (\log Q)^B$$

and PHNC supplies a fixed exponent $C = C(B)$, then [Theorem 1](#) gives

$$\mathcal{E}_* \gg_\alpha \frac{Q}{\ell^{C+3} \log Q} \gg_{\alpha, B, C} \frac{Q}{(\log Q)^{B(C+3)+1}}.$$

Thus $\mathcal{E}_* \rightarrow \infty$ faster than any negative power of $\log Q$, and the normalized collision excess in [\(1\)](#) tends to zero exponentially in this scale.

Remark 1 (What this does and does not replace). *This theorem shows that Paper 10's constant angular non-concentration hypothesis is stronger than needed for collision-excess decay. It does not prove PHNC. It also does not by itself replace KRWS or Kummer-BV in Paper 4: it supplies a Plancherel L^2 collision decay input, while KRWS and BVK still encode additional smoothing and sieve-transfer requirements.*

6 Rank- r status

The theorem is not special to $r = 2$. For rank r , the correct PHNC input is hyperplane non-concentration:

$$\#\{q \in \mathcal{Q} : x_q \notin H_A\} \geq \ell^{-C} |\mathcal{Q}| \quad (A \neq 0).$$

The only rank dependence in the collision bound is the harmless Plancherel prefactor $1 - \ell^{-r}$. For fixed r , this prefactor does not compete with the exponential decay. If r grows with Q , one must separately require $\mathcal{E}_* \gg r \log \ell$ to dominate the number of nonzero Fourier modes.

7 Dependency status

- The PHNC-to-angular bridge is finite once [Hypothesis 1](#) or [Hypothesis 2](#) is assumed.
- The proof uses only Paper 9’s angular-energy bound and the elementary cosine gap in [Lemma 1](#).
- The result is conditional on PHNC or weighted PHNC. It does not prove PHNC.
- The result proves collision-excess decay, not the full KRWS/BVK sieve conclusion.
- No Erdős–Graham #203 resolution is asserted.